RAG Learning

P1 — Matter

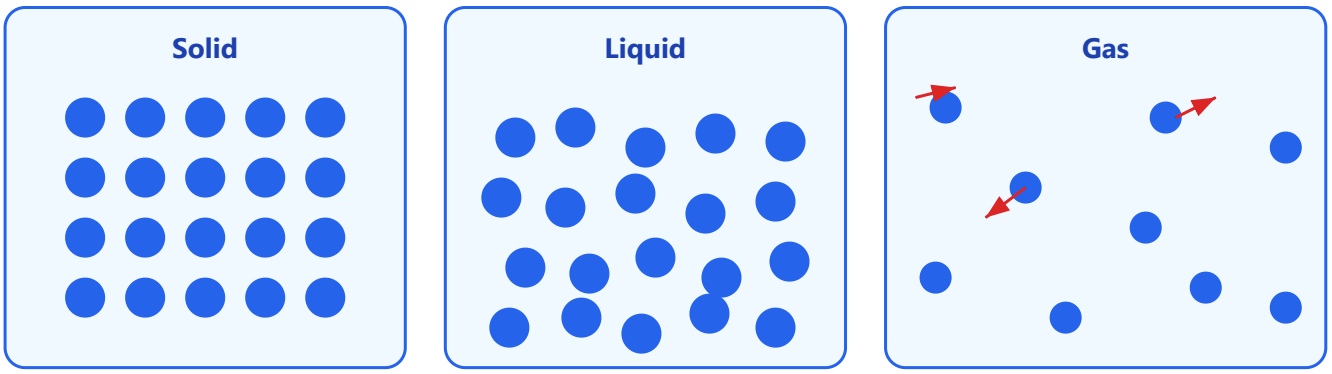
OCR GCSE Physics A (J249) — Gateway Science · Paper 1

P1.1 The Particle Model

> The **particle model** states that all matter is made of tiny particles (atoms or molecules) that are in constant motion. The **arrangement, spacing and motion** of these particles determines whether a substance is a solid, liquid or gas.

We cannot see atoms directly, but the model explains a huge range of observations — from why liquids flow to why balloons deflate. You need to know the typical particle arrangement in each state and how the particles behave.

Arrangement of Particles in the Three States of Matter



Regular pattern · touching · vibrate in place | Random · touching · slide past each other | Random · far apart · fast, random motion

Properties of the Three States

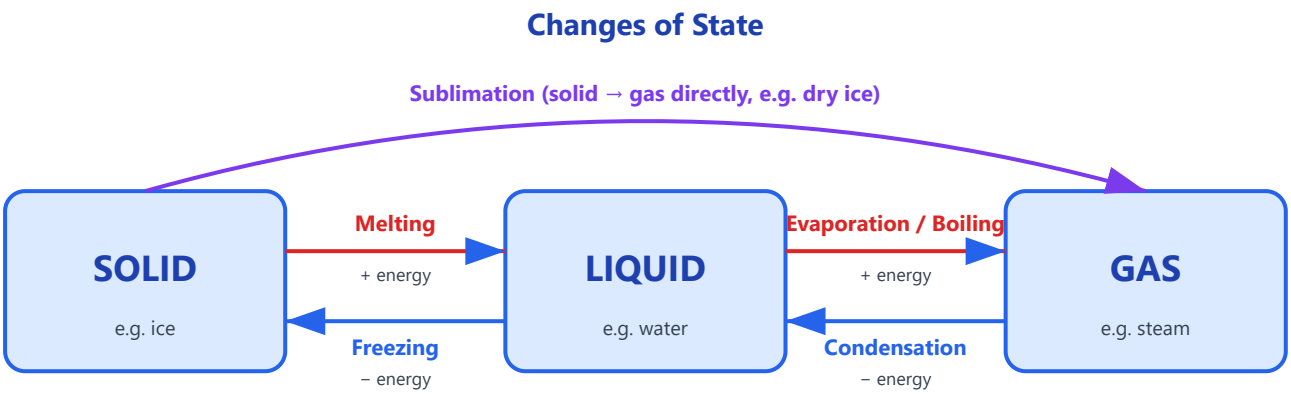
Property	Solid	Liquid	Gas
Shape	Fixed	Takes shape of container	Fills container

Volume	Fixed	Fixed	Fills container (can be compressed)
Compressible?	No	Barely	Yes — lots of empty space
Density	Usually high	Slightly lower than solid (water is unusual — ice is less dense)	Very low
Particle arrangement	Regular, tightly packed	Random but close together	Random and far apart
Particle motion	Vibrate about fixed positions	Slide past each other	Move quickly in all directions
Forces between particles	Strong	Moderate	Negligible (except during collisions)

Exam Tip: A common trap: gases *fill* their container, they do not just "take its shape". Because there are almost no intermolecular forces between gas particles, they spread out to occupy all the space available to them.

P1.2 Changes of State

A **change of state** is a physical change — the chemical composition does not alter, so the change is **reversible**. Energy must be transferred to or from the substance to break or form the forces between particles.



Mass is always conserved in a change of state — the same number of particles exists before and after, they are just rearranged. A gas condensing back to a liquid gives back exactly the same mass of liquid.

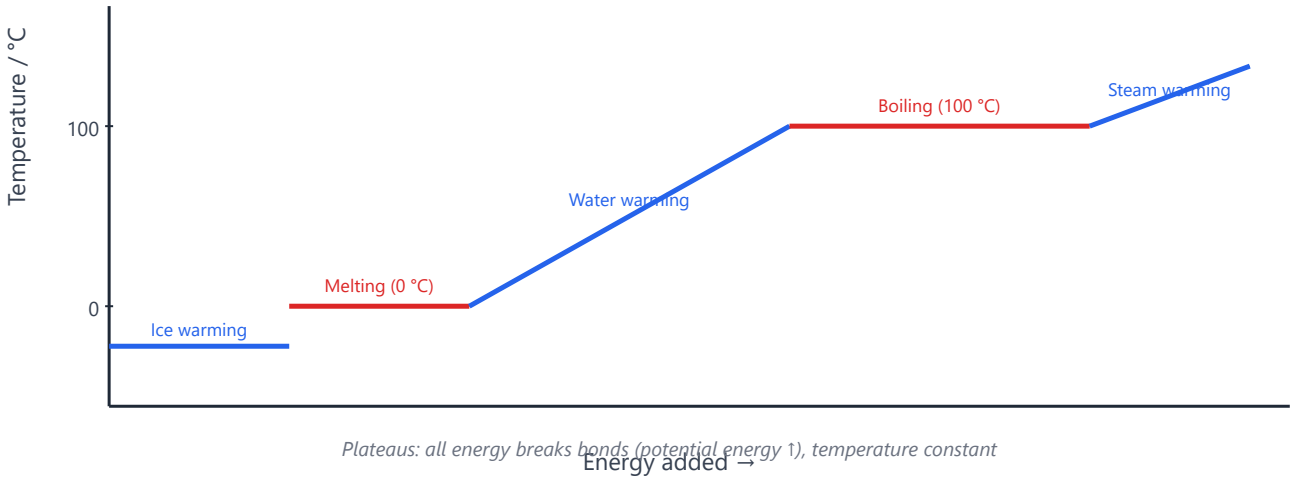
Internal Energy

> The **internal energy** of a system is the total energy stored by the particles. It is the sum of the **kinetic energy** of the particles (from their motion) and the **potential energy** stored in the bonds/intermolecular forces between them.

Heating a substance transfers energy to its particles. This energy either:

- Increases the **kinetic energy** of the particles — they move faster and the **temperature rises**, OR
- Increases the **potential energy** of the particles — bonds/forces between particles are overcome, causing a **change of state** (temperature stays constant during the change).

Heating Curve for Water (Temperature vs Energy Added)



Plateaus: all energy breaks bonds (potential energy ↑), temperature constant

Specific Heat Capacity

The **specific heat capacity (c)** of a substance is the amount of energy required to raise the temperature of **1 kg** of the substance by **1 °C**. Units: J/kg °C.

Energy transferred (ΔE) = mass (m) × specific heat capacity (c) × temperature change (Δθ)

$$\mathbf{\Delta E = m \times c \times \Delta\theta} \quad (\text{J}) = (\text{kg}) \times (\text{J/kg °C}) \times (\text{°C})$$

Worked Example: A 2 kg block of copper (c = 385 J/kg °C) is heated from 20 °C to 70 °C. Energy transferred?
ΔE = m × c × Δθ = 2 × 385 × (70 − 20) = 2 × 385 × 50 = **38 500 J = 38.5 kJ**.

Water has an unusually high specific heat capacity (c ≈ 4200 J/kg °C). That is why it is used in car radiators and central-heating systems — it can absorb and release lots of energy without a huge temperature change.

Specific Latent Heat

The **specific latent heat (L)** is the energy needed to change the state of 1 kg of a substance **without changing its temperature**. Units: J/kg.

- **Latent heat of fusion** — for melting / freezing.
- **Latent heat of vaporisation** — for boiling / condensing.

Energy for change of state = mass × specific latent heat

$$\mathbf{E = m \times L} \quad (\text{J}) = (\text{kg}) \times (\text{J/kg})$$

Worked Example: How much energy is needed to melt 0.5 kg of ice (L = 334 000 J/kg)?
E = m × L = 0.5 × 334 000 = **167 000 J = 167 kJ**.
Note: this is the energy used *only* during melting — raising the temperature of the resulting water is a separate calculation using c.

Exam Tip: If a question mixes warming *and* a change of state (e.g. ice at −10 °C → steam at 110 °C), split it into stages and use c for temperature-change stages and L for change-of-state stages. Add the answers together.

P1.3 Density

Density (ρ) is the mass per unit volume of a substance. It tells you how "heavy" a given volume of a material is.

$$\boldsymbol{\rho = m / V} \quad (\text{kg/m}^3) = (\text{kg}) / (\text{m}^3)$$

Also common: g/cm³. 1 g/cm³ = 1000 kg/m³.

Material	Density (kg/m³)	Why
Air (20 °C)	~1.2	Gas — particles far apart
Wood (oak)	~700	Contains air gaps; floats on water

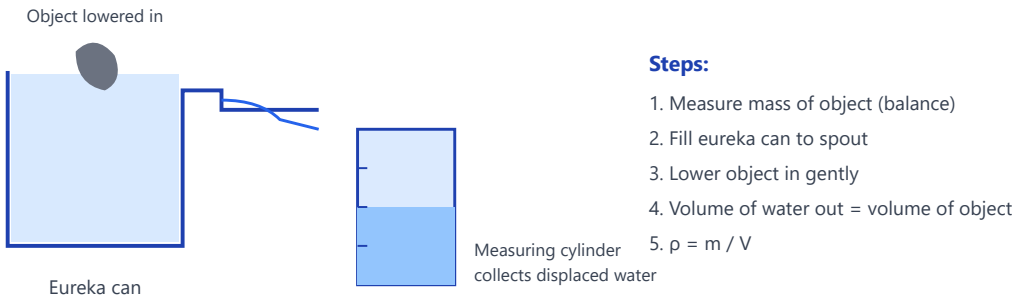
Water	1 000	Standard reference — 1 g/cm³
Aluminium	2 700	Light metal
Iron	7 870	Dense metal
Lead	11 340	Heavy; historic radiation shielding
Gold	19 300	Very dense — "heavy for its size"

Measuring Density

The method depends on whether the object is regular, irregular, or a liquid:

- **Regular solid** (cube, cylinder): measure mass with a balance. Calculate volume from dimensions with a ruler / vernier calliper ($V = l \times w \times h$ for a cuboid, $V = \pi r^2 h$ for a cylinder).
- **Irregular solid**: measure mass with a balance. Measure volume by **displacement** — lower the object into a partially-full measuring cylinder or eureka can. Volume displaced = volume of object.
- **Liquid**: place an empty measuring cylinder on a balance, zero it, pour in a known volume (e.g. 50 cm³), record mass. $\rho = m / V$.

Displacement method for irregular objects



Worked Example: A stone has mass 135 g and displaces 50 cm³ of water. Its density?
$\rho = m / V = 135 / 50 =$ **2.7 g/cm³ (= 2700 kg/m³)**. That's the density of aluminium / granite.

Why Density Varies Between States

The same substance in different states usually has very different densities because the **spacing between particles** changes. Solids and liquids of the same substance have similar densities because the particles are still touching. A gas has a much lower density because the particles are far apart.

Exam Tip: Water is unusual — its solid form (ice) is *less* dense than its liquid form, so ice floats. This is because water molecules form an open hexagonal lattice when they freeze.

P1.4 Pressure

Pressure is force per unit area — how much force is pushing on a given area. Unit: pascal (Pa) = 1 N/m^2.

$$\mathbf{p = F / A} \quad (\text{Pa}) = (\text{N}) / (\text{m}^2)$$

The same force produces a much higher pressure if applied over a smaller area. That's why a sharp knife cuts easily (force concentrated on a tiny area) but a snowshoe stops you sinking in snow (force spread over a large area).

Worked Example: A 600 N person stands on ice. With boots, their boots cover 0.015 m^2. With snowshoes, they cover 0.30 m^2. Compare the pressure they exert.

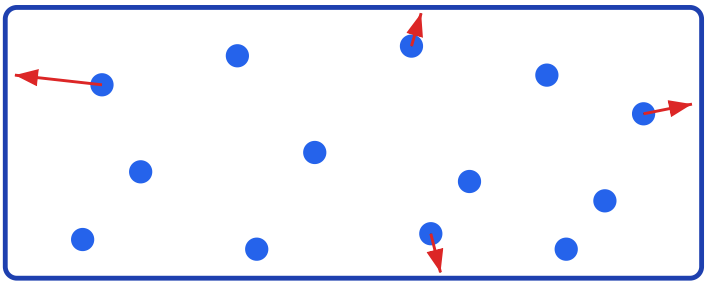
- Boots: p = 600 / 0.015 = **40 000 Pa**.
- Snowshoes: p = 600 / 0.30 = **2 000 Pa**.

The snowshoes reduce the pressure by a factor of 20 — enough to stop sinking.

Pressure in Gases

Gas particles move randomly at high speed and collide with the walls of their container. Each collision exerts a tiny force. The **total force per unit area from all these collisions** is the **gas pressure**.

Gas pressure arises from particle collisions with walls



Each collision exerts a force. Pressure = total force per unit area of container wall.

Effect of Temperature on Gas Pressure

When you **increase the temperature** of a gas (at constant volume):

1. Particles gain kinetic energy — they move **faster**.
2. They hit the walls **more often**.
3. They hit the walls **harder** (bigger change of momentum per collision).
4. Net result: **pressure increases**.

Doubling the absolute temperature (in kelvin) doubles the pressure, if volume is constant. This is why an aerosol can explodes if heated — the pressure inside rises above what the can can contain.

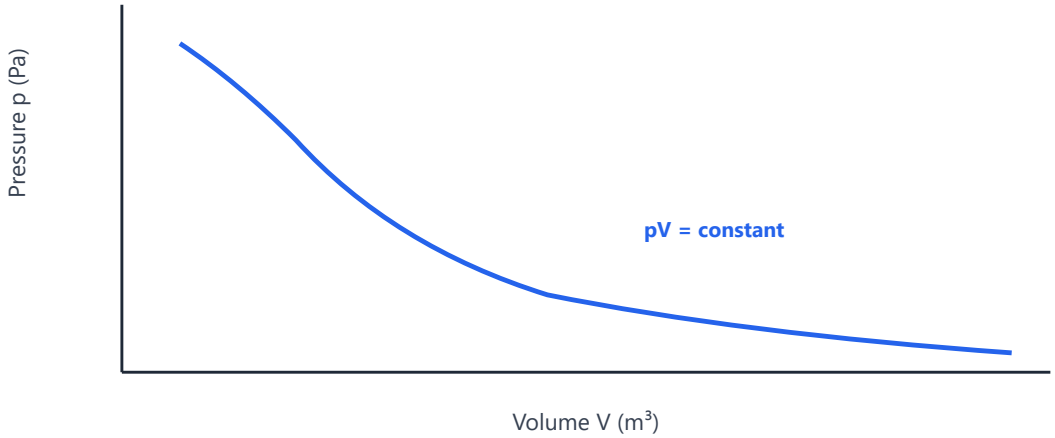
Gas Pressure and Volume (Boyle's Law)

At **constant temperature**, compressing a gas into a smaller volume increases its pressure. Halving the volume doubles the pressure, because the particles hit the walls twice as often.

$$\mathbf{p \times V = constant} \quad \text{(at constant temperature)}$$

$$\text{So: } p_1V_1 = p_2V_2$$

Pressure vs Volume at constant temperature



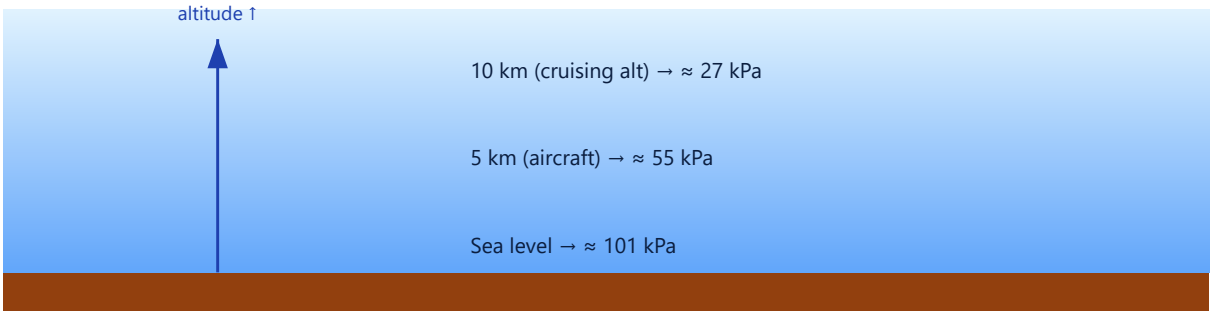
Worked Example: A gas has volume 2.0 m³ at pressure 100 kPa. It is compressed to 0.5 m³ (temperature unchanged). New pressure?

$p_1V_1 = p_2V_2 \Rightarrow 100 \times 2.0 = p_2 \times 0.5 \Rightarrow p_2 = 200 / 0.5 =$ **400 kPa**.

Atmospheric Pressure

The Earth's atmosphere has weight. The column of air pressing down on you creates **atmospheric pressure** — about 101 kPa (≈ 101 000 Pa) at sea level. Pressure **decreases with altitude** because there is less air above you.

Atmospheric pressure decreases with altitude



Less air above = fewer collisions = lower pressure

Pressure in Liquids

$$\mathbf{p = h \times \rho \times g} \quad \text{(Pa)}$$

h = height of liquid column (m) ρ = density of liquid (kg/m³) g = 9.8 N/kg

Pressure in a liquid acts in **all directions**, and it **increases with depth** because a deeper point has more liquid above it pressing down. Dams are built thicker at the base than the top for this reason.

Worked Example: Water is 3.0 m deep (ρ = 1000 kg/m^3). Pressure at the bottom due to the water?
$p = h \times \rho \times g = 3.0 \times 1000 \times 9.8$ = **29 400 Pa ≈ 29 kPa**. (Total pressure at bottom = this + atmospheric.)

Upthrust — Why Things Float

Because pressure increases with depth, the pressure on the bottom of a submerged object is greater than on the top. This difference produces a net upward force called **upthrust**, equal to the weight of fluid the object displaces (Archimedes' principle).

- If **upthrust > weight**: object floats and the object rises until enough is above the surface that upthrust = weight.
- If **upthrust = weight**: object floats in equilibrium at the current position.
- If **upthrust < weight**: object sinks.

A steel ship floats because its shape displaces a huge volume of water — enough that the weight of the displaced water exceeds the weight of the ship. A solid lump of steel of the same mass would not displace enough water and would sink.

Exam Tip: Know the three equations for pressure and when each applies:
• $\mathbf{p = F/A}$ — solid on a surface.
• $\mathbf{p = h\rho g}$ — liquid pressure at depth.
• $\mathbf{p_1V_1 = p_2V_2}$ — gas at constant temperature.
Check units carefully: volume in m^3 not cm^3, density in kg/m^3 not g/cm^3.

Quick Summary — Key Equations for P1

Quantity	Equation	Units

Density	$\rho = m / V$	kg/m^3
Pressure (solid)	$p = F / A$	Pa (= N/m^2)
Pressure in liquid	$p = h \times \rho \times g$	Pa
Boyle's Law	$p_1V_1 = p_2V_2$	—
Heating / cooling	$\Delta E = m \times c \times \Delta\theta$	J
Change of state	$E = m \times L$	J

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